

Notes: Benmelech, Guren & Melzer (RFS, 2023)

Making the House a Home: The Stimulative Effect of Home Purchases on Consumption and Investment

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1 Big picture

- **Question.** Why do household spending and housing market activity move together? The paper asks whether the act of buying a home itself creates demand for home-related durables, improvements, repairs, and maintenance.
- **New channel.** The paper defines the *home purchase channel*: households tailor newly purchased homes to their preferences by buying furniture/appliances and making repairs or improvements.
- **Main micro finding.** Around a primary home purchase, households spend about \$8,230 more on home-related durables and improvements from three months before through two years after the purchase.
- **Validation.** CE event studies, PSID move comparisons, and BuildFax permit data all point to a strong post-purchase spike in home-related spending.
- **Aggregate result.** In the Great Recession, the collapse in home sales explains roughly one-third of the decline in home durables and home improvement/maintenance spending.
- **Economic takeaway.** Housing market transaction volume is itself a macroeconomic transmission channel.

2 Data and measurement

2.1 Consumer Expenditure Survey (CE)

The CE provides monthly panel data from 2001 to 2013 and is the main source for the household event study. It measures home improvements, home durables, nonhome durables, nondurables, and services. The sample includes 571,871 monthly observations on 60,642 homeowners.

Interpretation: The CE is best suited for measuring the immediate timing of home-related consumption around the exact purchase month.

2.2 Panel Study of Income Dynamics (PSID)

The PSID provides a separate longitudinal sample of homeowners and renters. It measures home improvements, repairs, furnishings/equipment, nonhome goods and services, and whether the household moved since the prior interview.

Interpretation: The PSID is used to distinguish the effect of home purchase from the effect of moving residence.

2.3 BuildFax and DataQuick permit data

BuildFax provides property-level histories of residential building permits, and DataQuick provides property deeds. The data include permit incidence, number of permits, work type, and job cost in some jurisdictions.

Interpretation: Permit data validate survey results and allow the authors to check seller-side prepurchase improvements.

2.4 Economic Census and CBSA-level retail spending

The Economic Census reports retail sales in 2002, 2007, and 2012. The paper uses home-related retail categories and CoreLogic data on home purchases and house prices.

Interpretation: These data benchmark the home-purchase channel against housing wealth effects.

3 Models and empirical specifications

3.1 CE household event-study model

$$Spending_{it} = \delta_t + \theta_i + \sum_{m=-3}^{23} \beta_m 1\{MonthsSincePurchase_{it} = m\} + \gamma' X_{it} + \varepsilon_{it}. \quad (1)$$

Interpretation: The coefficients β_m measure within-household spending differences by month relative to the home purchase, net of time effects and controls.

3.2 PSID mover comparison

$$Spending_{it} = \delta_t + \theta_i + \beta 1\{MovedResidenceSincePriorInterview_{it}\} + \gamma' X_{it} + \varepsilon_{it}. \quad (2)$$

Interpretation: Estimating this separately for homeowners and renters tests whether the spending spike is caused by ownership/purchase rather than moving alone.

3.3 BuildFax property event-study model

$$Permits_{pt} = \delta_t + \theta_p + \sum_{q=-8}^8 \beta_q 1\{QuartersSincePurchase_{pt} = q\} + \gamma' X_{pt} + \varepsilon_{pt}. \quad (3)$$

Interpretation: This property-level event study uses administrative permit records to measure large home improvement activity before and after sale.

3.4 Aggregate contribution

$$ExplainedShare_{c,t_0 \rightarrow t_1} = \frac{\Delta HomeSales_{t_0 \rightarrow t_1} \times EstimatedSpendingPerPurchase_c}{\Delta AggregateSpending_{c,t_0 \rightarrow t_1}}.$$

Interpretation: This converts the household spending response into an aggregate boom/bust contribution.

3.5 CBSA retail-sales elasticity model

$$\Delta \log Spending_{ct} = \alpha + \beta \Delta \log HomePurchases_{c,t-1} + \gamma \Delta \log HomePriceIndex_{c,t-1} + \delta_t + \varepsilon_{ct}. \quad (4)$$

Interpretation: The coefficients compare the home purchase channel with the housing wealth channel in metropolitan-level retail data.

4 Identification and empirical design

The key identifying variation is within-household timing relative to purchase, with household fixed effects and month-by-year fixed effects. The category pattern is also important: home-related spending rises sharply, while nonhome durables and nondurables/services do not. PSID renters show that moving alone cannot explain the homebuyer response. BuildFax permit data validate large postpurchase improvements and provide a longer window around transactions.

5 Section-by-section study guide

- **Introduction.** Defines the home-purchase channel and explains why transaction volume may affect consumption separately from wealth.
- **Data and Methodology.** Introduces CE, PSID, BuildFax/DataQuick, and Economic Census data.
- **CE event studies.** Figures 2 and 3 estimate month-by-month responses of home durables and home improvements.
- **PSID comparison.** Table 5 compares homeowner movers and renter movers.
- **BuildFax analysis.** Figure 4 validates large home improvement activity with permit data.
- **Nonhome spending.** Figure 5 checks whether unrelated spending rises similarly.
- **Aggregate effect.** Figure 6 and Table 6 convert micro responses into Great Recession magnitudes.
- **CBSA elasticities.** Table 7 compares purchases and house prices as drivers of home-related retail spending.

6 Tables and figures

6.1 Figure 1: Home sales and home-related spending in the Great Recession

Read: Home sales and home-related spending are normalized to their own peaks. Both collapse during the housing bust.

Detailed interpretation: This is motivating aggregate evidence, not causal evidence. It shows the macro comovement the paper seeks to explain with microdata.

Economic message: Home transaction volume is a plausible driver of home-related consumption.

6.2 Table 1: Summary statistics for homeowners in Consumer Expenditure Survey

Read: Average monthly home improvement spending is \$222, and home durable spending is \$114. Home improvement occurs in 18% of months and home durable purchases in 29% of months.

Detailed interpretation: The table establishes that home-related spending is common, economically meaningful, and lumpy.

Economic message: The CE is appropriate for monthly event-study estimates around home purchases.



Figure 1
Home sales and home-related spending in the Great Recession
This figure illustrates the time patterns in home sales and home durables, improvements, and maintenance spending between 2001 and 2014. Both series are normalized by their maximum value. Home sales (in blue and scaled to the left axis) are the sum of the National Association of Realtors' existing home sales series and the Census's series of new home sales. A 12-month moving average centered on the indicated date is shown to smooth out seasonality. Home durables, improvement, and maintenance (in red and scaled to the right axis) is the sum of these categories from the Consumer Expenditure Survey aggregated up by the survey weights and normalized to US\$(2009) using the category price index.

Table 1
Summary statistics for homeowners in Consumer Expenditure Survey

A. Household spending			C. Household characteristics		
	Mean	SD		Mean	SD
Spending per month (\$)			Income/wealth		
Home improvement	222	2,071	Annual income	77,492	68,610
Home durables	114	739	Financial assets	62,766	310,118
Non-home durables	536	3,617	Missing assets?	0.11	0.32
Non-durables and services	2,643	2,199	Education		
Likelihood of purchase (per month)			No HS diploma	0.10	0.30
Any home improvement?	0.18	0.39	HS diploma	0.25	0.43
Any home durables?	0.29	0.45	Some college	0.29	0.46
Any non-home durables?	0.53	0.50	College degree	0.22	0.42
Any durables or improvement?	0.65	0.48	Graduate degree	0.14	0.34
B. Property characteristics			Race/ethnicity		
	Mean	SD	White	0.72	0.45
Months since purchase	157	154	Black	0.09	0.28
Purchased past 12 mo? (%)	6.66	24.94	Hispanic	0.14	0.35
Mortgagor? (%)	66.43	47.22	Asian	0.04	0.19
Months since refinancing	58	69	Marital status		
Refinanced prior 12 mo? (%)	7.18	25.82	Married	0.64	0.48
Age of home (years)	36	29	Widowed	0.10	0.30
Age of home missing?	0.09	0.28	Divorced	0.13	0.33
Rooms	6.67	2.06	Separated	0.01	0.12
Bedrooms	3.13	0.90	Never married	0.11	0.31
Bathrooms	1.83	0.74	Other		
Central air?	0.70	0.46	Age (HH head)	52.57	15.74
Swimming pool?	0.12	0.32	Family size	2.66	1.46
Porch?	0.82	0.38	Retired? (%)	0.23	0.42
Off-street parking?	0.83	0.38			

This table presents summary statistics for homeowners interviewed in the Consumer Expenditure Survey between 2001 and 2013. The sample includes 571,871 monthly observations on 60,642 households. The sample excludes homeowners who do not report their date of home purchase.

6.3 Tables 2 and 3: PSID and BuildFax summary statistics

Read: PSID homeowners spend more than renters on home-related categories, and renters move more often. BuildFax permits are rare but large when they occur.

Detailed interpretation: PSID helps separate moving from home purchase; BuildFax captures large improvements that surveys may miss.

Economic message: Multiple datasets address different threats to interpretation.

6.4 Table 4: Household spending and building permit activity following a home purchase

Read: First-quarter spending after purchase is high: \$615 in improvements and \$798 in home durables in CE data. Permit incidence is 7.7% in the first quarter and then declines.

Detailed interpretation: The raw patterns show a postpurchase spike before controls and fixed effects are applied.

Economic message: Home purchase is associated with immediate home-related spending and large home-improvement projects.

6.5 Figures 2 and 3: Home durables and home improvement impulse responses

Read: Figure 2 shows home durable spending spikes in months 0–2 and fades by the end of the first year. Figure 3 shows improvement/maintenance spending rises after purchase and remains elevated longer.

Detailed interpretation: Durables are front-loaded; improvements persist because projects take time.

Economic message: Home purchase stimulates both consumption of durables and residential investment.

Table 2
Summary statistics for Panel Study of Income Dynamics

	Mean	SD	N
<i>A. Homeowners</i>			
Household spending (\$)			
Home improvements (past 2 years)	5,067	27,109	30,125
Home repairs (past year)	2,891	13,690	22,132
Home furnishings and equipment (past year)	1,402	5,271	22,132
Nonhome goods and services (past year)	30,314	22,420	22,132
Moving			
Moved residence since prior interview (%)	14	35	30,125
<i>B. Renters</i>			
Household spending (\$)			
Home improvements (past 2 years)	316	8,661	18,278
Home repairs (past year)	264	2,370	14,171
Home furnishings and equipment (past year)	586	1,871	14,171
Nonhome goods and services (past year)	18,105	14,839	14,171
Moving			
Moved residence since prior interview (%)	56	50	18,278

This table presents summary statistics of homeowners (panel A) and renters (panel B) surveyed in the Panel Study of Income Dynamics between 2001 and 2013. The sample includes 48,403 observations from biannual interviews of 10,358 unique households. We classify homeownership as of the interview date. The survey measures cumulative spending on home improvements over a roughly 2-year period preceding the interview. It measures cumulative spending on home repairs, furnishings, equipment, and nonhome goods and services over a 1-year period preceding the interview. Home improvement spending is available for the full sample period, while spending in other categories is available from 2005 to 2013.

Table 3
Summary statistics for building permits

	Mean	SD
<i>A. Building permits</i>		
Estimated total job cost (\$)		
Unconditional, per quarter	749	14,928
Conditional on permit	42,045	103,780
Number of permits, by type (per property)		
Electrical	0.16	1.10
Mechanical	0.10	0.75
Plumbing	0.10	0.78
Structural	0.53	2.64
All	0.88	4.23
<i>B. Property characteristics</i>		
Mortgagor? (%)	72.3	44.7
Age of home (years)	35.8	27.5
Bedrooms	3.2	0.8
Bathrooms	2.1	0.9
Transaction price (\$US(2009))	285,526	249,892
Foreclosure or short sale (%)	19.7	39.8
Change in log CBSA-level housing price index in 3 years (%)	19.5	26.9

This table presents summary statistics of building permits and property characteristics for homes sold between 2001 and 2013. The sample includes 19,727,786 property-quarter observations on 1,287,725 transactions and 967,904 unique properties. The source of the permit data in panel A is BuildFax, and the source of the property characteristics in panel B is DataQuick.

Table 4
Household spending and building permit activity following a home purchase

A. Spending after home purchase: Consumer Expenditure Survey

Quarters since purchase	Average monthly spending (\$)			
	Home improvement	Home durables	Nonhome durables	Nondurables and services
1	615	798	788	2,531
2	399	265	702	2,473
3	338	178	659	2,486
4	297	158	648	2,502
5+	209	97	525	2,656

B. Spending by homeowners after moving: Panel Study of Income Dynamics

Moved in prior 2 years?	Average spending (\$)			
	Home improvement (past 2 years)	Home durables (past year)	Home repairs (past year)	Nonhome goods and services (past year)
Yes	8,887	2,709	3,654	30,614
No	4,485	1,196	2,771	30,185

C. Building permits after home purchase: BuildFax-DataQuick sample

Quarters since purchase	Building permit activity		
	Any permit?	Number of permits	Estimated job cost (\$)
1	0.077	0.12	1,292
2	0.057	0.09	1,344
3	0.040	0.06	1,032
4	0.034	0.05	813
5+	0.026	0.04	506

This table summarizes homeowners' spending and building permit activity after they purchase a home. In panel A, we report average monthly spending for homeowners in the Consumer Expenditures Survey between 2001 and 2013. In panel B, we report average biannual or annual spending for homeowners in the Panel Study of Income Dynamics between 2001 and 2013. In panel C, we report the incidence of building permits and the average estimated job cost for home purchases in the BuildFax-DataQuick sample between 2001 and 2013. We measure spending and job cost in real terms, using the CPI-U as price deflator and January 2009 as the base period.

6.6 Table 5: Household spending by homebuyers and renters that move

Read: Homebuyer movers spend \$6,227 more on improvements, \$1,916 more on repairs/maintenance, and \$1,924 more on furnishings/equipment/appliances. Renter movers show only modest increases.

Detailed interpretation: The homebuyer-renter contrast separates purchase/ownership from the act of moving.

Economic message: The channel is tied to tailoring an owned home, not relocation alone.

6.7 Figure 4: Building permit impulse response to home purchase

Read: Permit job costs and permit incidence rise around and after home purchase; there is also some prepurchase seller activity.

Detailed interpretation: Permit data support the CE results and show large projects, not just small durable purchases.

Economic message: Home purchase triggers residential investment that may affect the quality of the housing stock.

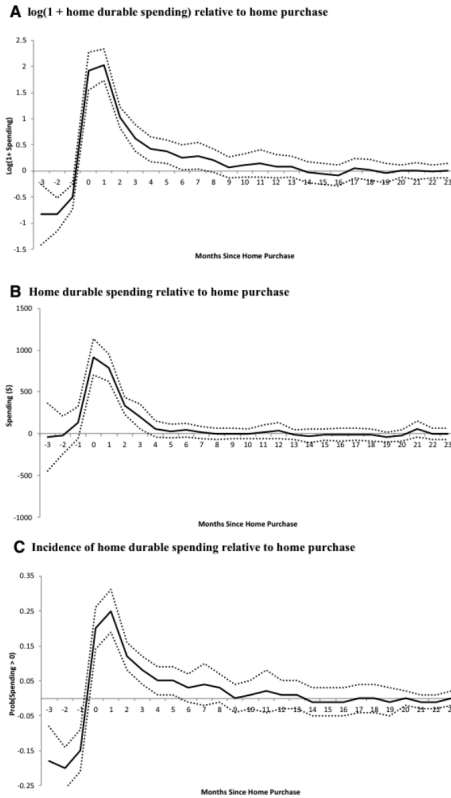


Figure 2
Home durables impulse response to home purchase
 The panels show the time pattern in homebuyers' home durable spending from 3 months before to 2 years after their home purchase. In each panel, we plot the coefficient estimates (solid line) and 95% confidence interval bounds (dotted lines) from estimating Equation (1) in the CE sample. Only the functional form of the spending measure varies across the three panels. Each model includes controls for property and household characteristics as well as household and year-by-month fixed effects. We estimate the models by ordinary least squares and two-way cluster observations by state and year-month in calculating standard errors.

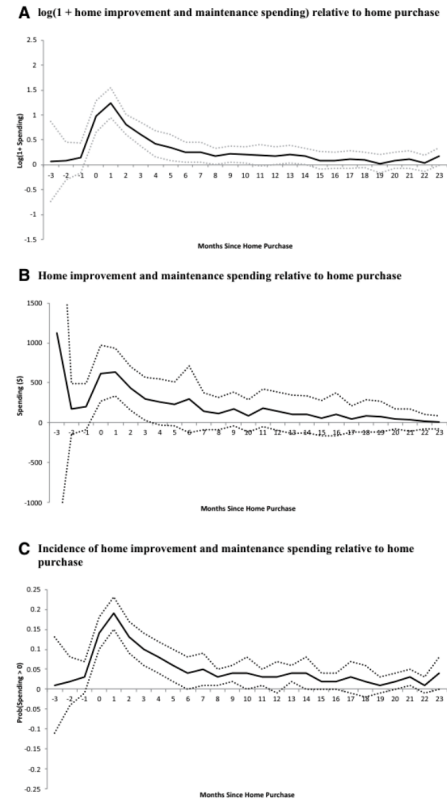


Figure 3
Home improvement and maintenance impulse response to home purchase
 The panels show the time pattern in homebuyers' home improvement and maintenance spending from 3 months before to 2 years after their home purchase. In each panel, we plot the coefficient estimates (solid line) and 95% confidence interval bounds (dotted lines) from estimating Equation (1) in the CE sample. Only the functional form of the spending measure varies across the three panels. Each model includes controls for property and household characteristics as well as household and year-by-month fixed effects. We estimate each model by ordinary least squares and two-way cluster observations by state and year-month in calculating standard errors.

Table 5
Household spending by homebuyers and renters that move (PSID)

	Home improvement	Home repair and maintenance	Home furnishings, equipment, and appliances	Nonhome expenditures
<i>A. Homebuyers</i>				
Moved residences?	6,227*** (1364)	1,916* (741)	1,924*** (309)	-421 (746)
N	29,316	21,125	21,125	21,125
R ²	.33	.27	.28	.72
<i>Control variables</i>				
Property characteristics?	Y	Y	Y	Y
HH characteristics?	Y	Y	Y	Y
HH fixed effects?	Y	Y	Y	Y
Year fixed effects?	Y	Y	Y	Y
<i>B. Renters</i>				
Moved residences?	42 (148)	98 (77)	238** (61)	292 (427)
N	17,145	13,008	13,008	13,008
R ²	.64	.40	.38	.69
<i>Control variables</i>				
Property characteristics?	Y	Y	Y	Y
HH characteristics?	Y	Y	Y	Y
HH fixed effects?	Y	Y	Y	Y
Year fixed effects?	Y	Y	Y	Y

This table presents regression analysis of home-related spending by homebuyers and renters in the Panel Study of Income Dynamics (PSID) between 2001 and 2013. We regress real spending (measured in US\$(2009)) on control variables and an indicator for whether the household moved residences during the prior 2-year period. The control variables are year fixed effects, household fixed effects, the number of rooms in the residence, log household income, log household size, household wealth (transformed by inverse hyperbolic sine), and indicators for the marital status, retirement status, education and race of the household head. Home improvement spending is measured over the 2-year period prior to the household interview, while the remaining home-related spending is measured over a 1-year period prior to the interview. We estimate the model using ordinary least squares and double-cluster observations by the state of residence and year in the calculation of standard errors. * $p < .1$; ** $p < .05$; *** $p < .01$.

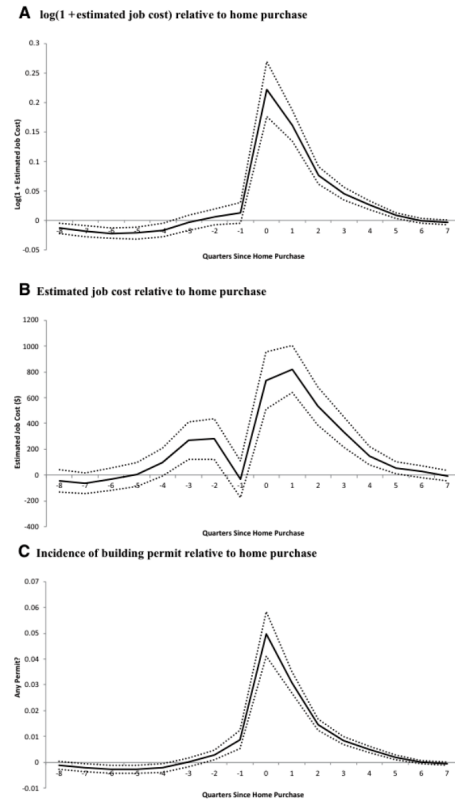


Figure 4
Building permit impulse response to home purchase
The panels show the time pattern in homebuyers' building permitting activity from 2 years before to 2 years after their home purchase. In each panel, we plot the coefficient estimates (solid line) and 95% confidence interval bounds (dotted lines) from estimating Equation (3) in the BuildFax-DataQuick sample. The dependent variable is the estimated cost of building permits, in logs or levels, in the first two panels and an indicator for any building permit in the third panel. Each model includes controls for property characteristics, as well as property and year-by-quarter fixed effects. We estimate each model by ordinary least squares and two-way cluster observations by CBSA and year-quarter in calculating standard errors.

6.8 Figure 5: Nonhome durables and nondurables/services responses

Read: Nonhome categories do not rise like home-related categories; some decline modestly around purchase.

Detailed interpretation: This placebo-style evidence weakens the omitted-wealth-shock interpretation.

Economic message: The response is home-specific.

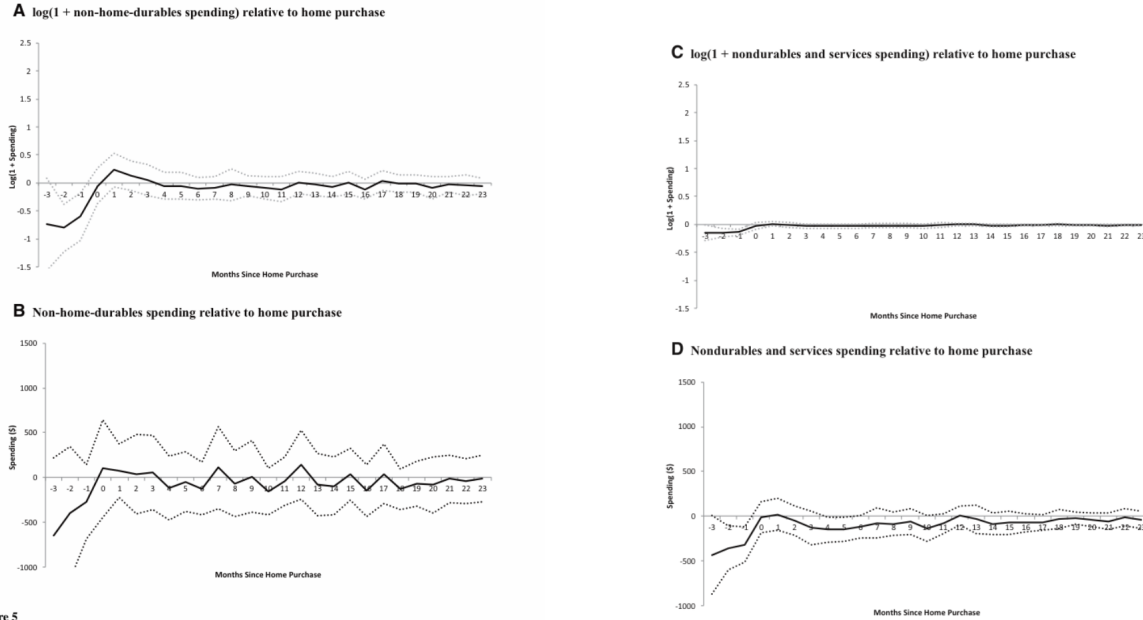


Figure 5
Nonhome durables and nondurables and services responses to home purchase
The panels show the time pattern in homebuyers' spending on non-home durables (panels A and B) and nondurables and services (panels C and D) from 3 months before to 2 years after their home purchase. In each panel, we plot the coefficient estimates (solid line) and 95% confidence interval bounds (dotted lines) from estimating Equation (1) in the CE sample. Each model includes controls for property and household characteristics as well as household and year-by-month fixed effects. We estimate each model by ordinary least squares and two-way cluster observations by state and year-month in calculating standard errors.

Figure 5
(Continued)

6.9 Figure 6: Cumulative spending relative to home purchase

Read: Cumulative home durables rise by about \$2,450; home improvements by about \$5,780; nonhome categories fall by about \$4,530; net spending rises by about \$3,700.

Detailed interpretation: The figure integrates monthly responses and shows that home-related increases dominate nonhome declines.

Economic message: Home purchases generate a net spending stimulus.

6.10 Table 6: Aggregate effect of home purchase channel in Great Recession

Read: During 2005–2010, the channel accounts for 30.7% of the decline in home durables and 31.5% of the decline in home improvements/maintenance.

Detailed interpretation: The micro response becomes macro-relevant because home sales collapsed sharply.

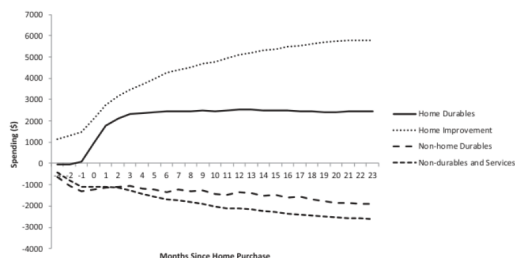


Figure 6
Cumulative spending relative to home purchase
The figure plots homebuyers' estimated cumulative spending relative to the month of their home purchase. We calculate cumulative spending by adding the monthly spending responses estimated from Equation (1) and reported in Figures 2 (panel B), 3 (panel B), 5 (panel B), and 5 (panel D).

Table 6
Aggregate effect of home purchase channel in Great Recession

	Home durables (%)	Home improvements and maintenance (%)	Nonhome durables (%)	Total durables, improvements, and maintenance (%)
2001–2005	7.8	19.4	0.3	7.9
2002–2005	8.0	25.6	0.5	10.1
2005–2008	37.2	43.1	0.3	17.2
2005–2009	21.5	33.5	0.8	16.4
2005–2010	30.7	31.5	0.2	13.5
2005–2011	36.2	33.7	0.6	16.2

Each cell reflects the fraction of the total change in a consumption category for a given time period explained by the home purchase channel, computed by multiplying the change in home sales for the time period by our preferred estimate of the dollar amount of consumption associated with a home purchase. For the preferred estimate and the consumption time series used in the denominator, the CE data are deflated by a category-level deflator from the CPI (for durables) or NIPA (for home improvements as well as maintenance, which does not have its own deflator). This is done at the monthly level and aggregated to the annual level. Each column reflects a consumption category, while each row reflects the time period over which changes are computed. The data on the total change in a consumption category for a given time period are aggregates computed with the CE data using CE sample weights. The sales series is created by combining National Association of Realtors (NAR) data on existing home sales with Census data on sales of new single-family homes, both taken from FRED (note that the NAR data are no longer on FRED but can be obtained from the NAR).

Economic message: The transaction-volume channel explains a large share of home-related spending declines.

6.11 Table 7: Elasticities of home-related retail spending to home purchases and home prices

Read: Home-purchase elasticities are 0.26 in 2002–2007 and 0.12 in 2007–2012. Home-price elasticities are 0.29 and 0.23.

Detailed interpretation: Purchases matter even controlling for prices; prices and transactions are complementary channels.

Economic message: Housing-cycle analysis should include both price and volume effects.

Table 7
Elasticities of home-related retail spending to home purchases and home prices

Time period:	Dependent variable: $\Delta \log (\text{Home-related retail spending})$	
	2002–2007	2007–2012
$\Delta \log (\text{home purchases})$	0.26*** (0.06)	0.12*** (0.02)
$\Delta \log (\text{home price index})$	0.29*** (0.03)	0.23*** (0.04)
Year fixed effects?	Y	Y
N	246	267
R^2	.44	.28

This table reports jointly estimated elasticities of home-related spending to home purchases and home prices. The spending data come from the 2002, 2007, and 2012 Economic Censuses, which report annual retail spending at all home improvement and home furnishing stores within a metropolitan area. The data on home purchases and home prices, which are also at the metropolitan area level, come from CoreLogic. We measure real spending and real home prices in 2012 dollars, using the CPI-U as the price deflator. We estimate the model with ordinary least squares, weighted by population, and cluster observations by metropolitan area when calculating standard errors, which are reported in parentheses. *** $p < .01$.

Table 7: Elasticities of home-related retail spending

7 Short summary

- The paper identifies a home-purchase channel linking home sales to spending.
- Homebuyers spend about \$8,230 more on home-related durables and improvements around purchase.
- PSID and BuildFax evidence validate the CE findings.
- Nonhome spending does not rise similarly, supporting a home-specific mechanism.

- The Great Recession collapse in sales explains roughly one-third of the fall in home-related spending categories.
- CBSA-level estimates show purchases and prices both predict home-related retail spending.

8 Conclusion

8.1 Core conclusion

Home purchases generate economically meaningful spending on home-specific goods and improvements. The channel is distinct from housing wealth effects and operates through transaction volume.

8.2 Economic intuition

A newly purchased home rarely matches the buyer’s preferences and existing durable stock. The purchase event triggers tailoring of the home through renovations, repairs, furniture, appliances, and maintenance.

8.3 Identification conclusion

The evidence is strongest because it combines within-household timing, renter comparisons, administrative permit data, and nonhome placebo categories.

8.4 Macro and policy implication

Monetary and fiscal policies that change home sales volume affect home-related consumption and investment directly. Transaction taxes or homebuyer subsidies may affect local retail and remodeling sectors even without price changes.

8.5 Limitations

Home purchases are not randomized; survey data have limited prepurchase windows; permits cover only permitted work; aggregate calculations are partial equilibrium.

8.6 Takeaway

Housing market fluctuations transmit through both prices and quantities. Ignoring transaction volume misses an important component of housing-cycle macroeconomics.